

BAYESUVIUS

A VISUAL DICTIONARY OF BAYESIAN
NETWORKS AND CAUSAL INFERENCE



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Bayesuvius,
a visual dictionary of Bayesian Networks and
Causal Inference

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This book is constantly being expanded and improved. To download the latest version, go to <https://github.com/rrtucci/Bayesuvius>

Bayesuvius

by Robert R. Tucci

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Figure 1: View of Mount Vesuvius from Pompeii



Figure 2: Mount Vesuvius and Bay of Naples

Contents

Foreword	22
Prologues	23
A Navigating the ocean of Judea Pearl’s Books	24
B CI-2-3 track	25
C Genomics Vocabulary	29
D Notational Conventions and Preliminaries	31
D.1 Some abbreviations frequently used throughout this book	31
D.2 Drawing Bayesian Networks	31
D.3 $\mathcal{N}(!a)$	32
D.4 Indicator function (a.k.a. Truth function)	32
D.5 One hot vector	32
D.6 L^p norm	32
D.7 Special sets	33
D.8 Kronecker delta function	33
D.9 Dirac delta function	34
D.10 Majority function	34
D.11 Underlined letters indicate random variables	34
D.12 Probability distributions	35
D.13 Independence, $\perp_P$	35
D.14 Discretization of continuous probability distributions	36
D.15 Samples, i.i.d. variables	37
D.16 Expected Value and Variance	38
D.17 Conditional Expected Value	38
D.18 $E[\underline{x}]$ from $P(\underline{x} < x)$	38
D.19 Notation for covariances	39
D.20 Conditional Covariance	40
D.21 Normal Distribution	41
D.22 Uniform Distribution	43
D.23 Softmax function (a.k.a. Boltzmann Distribution)	43
D.24 Sigmoid and log-odds functions	44

D.25	Estimand, Estimator (curve-fit), Estimate, Bias	45
D.26	Maximum Likelihood Estimate, Likelihood Ratio Test	45
D.27	Mean Square Error (MSE)	46
D.28	Cramer-Rao Bound	48
D.29	Bayes Rule, Bayesian Updating And Conjugate Priors	52
D.30	Entropy, Kullback-Leibler divergence, Cross-Entropy	54
D.31	Definition of various entropies used in Shannon Information Theory	55
D.32	Mean log likelihood asymptotic behavior	56
D.33	Arc Strength (Arc Force)	58
D.34	Pearson Chi-Squared Test	58
D.35	Demystifying Population and Sample Variances	59
D.36	Independence of $\hat{\mu}$ and $\hat{\sigma}^2$	60
D.37	Chi-square distribution	62
D.38	Student's t-distribution	62
D.39	Hypothesis testing and 3 classic test statistics (Likelihood, Score, Wald)	65
D.40	Error Bars	69
D.41	Confidence Interval	70
D.42	Score p-value	72
D.43	Convex/Concave functions, Jensen's Inequality	74
D.44	Chebyshev's inequality	75
D.45	Short Summary of Boolean Algebra	76
D.46	Laplace transform	78
	D.46.1 Examples	79
	D.46.2 Properties	80
D.47	Z-transform	86
	D.47.1 Examples	89
	D.47.2 Properties	91
D.48	Legendre Transformation (dual functions)	93
	D.48.1 Examples	94
	D.48.2 Properties	97
	D.48.3 Connection to Fourier transform and Quantum Mechanics	99
D.49	Numpy tensor methods	100
E	Linear Regression, Ordinary Least Squares (OLS)	106
E.1	LR, assuming x_σ are non-random	107
	E.1.1 Derivation of LR From Minimization of Error	107
	E.1.2 Geometry of LR with non-random x_σ	108
	E.1.3 LR Goodness of Fit, R^2	110
E.2	LR, assuming x_σ are random	112
	E.2.1 Transforming from non-random to random x_σ	113
	E.2.2 LR with random x_σ , expressed in derivative notation	115

E.2.3	R^2 with random x_σ	119
E.2.4	Regressing Residuals	120
E.3	Logistic Regression (LoR)	121
F	Classic Bayesian Network Apps	123
G	Definition of a Bayesian Network	127
H	Bayesian Networks, Causality and the Passage of Time	131
H.1	Unifying Principle of this book	131
H.2	You say tomato, I say tomato	132
H.3	A dataset is causal model free	132
H.4	What is causality?	133
H.5	Bayesian Networks and the passage of time	134
H.6	Advice for the DAG-phobic	135
1	A/B testing	136
1.1	Simple Example of A/B testing	136
1.2	Statistics formulae	136
2	AdaBoost	141
2.1	AdaBoost for general ensemble of w-classifiers	141
2.2	AdaBoost for ensemble of tree stumps	145
3	ANOVA	147
3.1	Introduction	147
3.2	Law of Total Variance	147
3.3	Sum of Squares Estimates	148
3.4	F-statistic and hypothesis testing	151
4	ARACNE structure learning	153
5	Autoregulon Networks (Network Motifs)	155
5.1	Hill functions	155
5.2	One Autoregulon	156
5.2.1	Autoregulon notation	158
5.2.2	Potential function	159
5.2.3	Linear Expansion of h	162
5.2.4	Lowpass h	163
5.2.5	Highpass h	164
5.2.6	Pulsed Lowpass h	166
5.2.7	Time dependent production $h(t)$	166
5.2.8	Fixed Points	167
5.3	Two autoregulons	168

5.3.1	Linearized $(+)(-)$	168
5.3.2	Linearized $(-)^2$ and $(+)^2$	169
5.3.3	Genetic Toggle Switch	170
5.4	More than 2 autoregulons	172
5.4.1	$(-)^2$ and $(+)^2$ Regulator nets	172
5.4.2	Regulated $(-)^2$ and $(+)^2$ nets	173
5.4.3	Feed Forward Loops	173
5.4.4	C1-FFL	175
5.4.5	I1-FFL	177
5.4.6	Chain of autoregulons	179
5.4.7	Repressilator	181
5.4.8	SIM net	183
5.4.9	Flagellum net	184
5.4.10	Bacillus Subtilis Sporation net	186
5.4.11	Frog Egg Cell Cycle net	186
5.5	Biology behind autoregulon net math	187
5.6	Summary of Network Motifs	190
6	Backdoor Adjustment Formula	192
6.1	Examples	192
7	Back Propagation (Automatic Differentiation)	197
7.1	Toy Example	197
7.2	General Theory	198
7.2.1	Jacobians	198
7.2.2	Bnets for function composition, forward propagation and back propagation	199
7.3	Application to Neural Networks	201
7.3.1	Absorbing b_i^λ into w_{ij}	201
7.3.2	Bnets for function composition, forward propagation and back propagation for NN	202
7.4	General bnets instead of Markov chains induced by layered structure of NNs	205
8	Bell and Clauser-Horne Inequalities in Quantum Mechanics	206
9	Berkson's Paradox	207
10	Binary Decision Diagrams	209
10.0.1	Conversion of Binary Tree into BDD	211
10.1	Equivalent Bnet	211
11	Chow-Liu Trees and Tree Augmented Naive Bayes (TAN)	214

11.1	Chow-Liu Trees	214
11.2	Tree Augmented Naive Bayes (TAN)	218
12	Control Theory (linear, deterministic)	220
12.1	Basic feedback model	221
12.2	Classical model (analog)	222
12.3	Modern model (analog)	225
12.4	Classical model (digital)	229
12.5	Modern model (digital)	229
12.5.1	Discretizing derivatives	229
12.5.2	Solving Difference Equation	231
12.6	Higher than first order differential (or difference) equations	233
12.6.1	Differential Equations	233
12.6.2	Difference Equations	234
12.7	Time-Invariance, Causality, Stability	234
12.8	Controllability, Observability	236
12.9	Signal Flow Graph	236
13	Copula	241
13.1	Examples	244
14	Counterfactual Reasoning	247
14.1	The 3 Rungs of Causal AI	247
14.2	Do operator	248
14.3	Imagine operator	248
15	Cross-Validation	252
16	DAG Extraction From Text (DEFT) or Time-Series	255
17	Dataset Shift and Batch Normalization	256
17.1	Covariate Shift	257
17.2	Concept Shift	257
17.3	Batch Normalization	258
18	Decision Trees	259
18.1	Conversion to Bnet	260
18.2	Structure Learning for Dtrees	261
18.2.1	Information Gain, Gini	262
18.2.2	Pseudocode	265
19	Decisions Based on Rungs 2 and 3: COMING SOON	267
20	Difference-in-Differences	268

20.1	John Snow, DID and a cholera transmission pathway	268
20.2	PO analysis	270
20.3	Linear Regression	272
21	Diffusion Models	275
21.1	Bnet for DM	275
21.2	Mean Values $M^{t-1}(x^t)$ and $M_\theta^{t-1}(x^t)$	278
21.3	Loss function $\mathcal{L}$	281
21.4	Algorithms for training and sampling DM	283
22	Digital Circuits	284
22.1	Mapping any dcircuit to a bnet	284
22.1.1	Option A of Fig.22.2	284
22.1.2	Option B of Fig.22.2	285
23	Dimensionality Reduction	286
24	Do Calculus	288
24.1	Observed and Hidden Nodes	291
24.2	3 Rules of Do Calculus	291
24.3	Parent Adjustment Formula	295
24.4	Backdoor Adjustment Formula	295
24.5	Frontdoor Adjustment Formula	297
24.6	Comparison of Backdoor and Frontdoor adjustment formulae	297
24.7	Do operator for DEN bnets	298
25	Do Calculus proofs	301
26	D-Separation	319
27	D-Separation in Quantum Mechanics	324
28	Dynamical Bayesian Networks	325
29	Dynamical Systems	327
29.1	Potential Function	328
29.2	Phase Portraits	329
29.3	Fixed Points in 1D Systems	331
29.3.1	Linear System	331
29.3.2	Quadratic System	331
29.3.3	Cubic System	332
29.4	Fixed Points in 2D Linear Systems	333
29.4.1	Potential function	336
29.4.2	Classification of fixed points	337

29.5	Bifurcations	338
29.5.1	Saddlepoint Bifurcation	339
29.5.2	Transcritical Bifurcation	339
29.5.3	Supercritical Pitchfork Bifurcation	340
29.5.4	Subcritical Pitchfork Bifurcation	340
29.5.5	Hopf Bifurcation	341
29.6	Hysteresis	343
29.7	Limit cycles	344
29.8	Homoclinic and Heteroclinic orbits	345
29.9	Oscillators	346
29.9.1	Linear oscillator	346
29.9.2	Nonlinear Oscillators	350
29.10	Chaos	353
29.11	Famous Models	353
29.11.1	Classical Mechanics	354
29.11.2	FitzHugh-Nagumo model	355
29.11.3	Heteroclinic Orbit Model	356
29.11.4	Homoclinic Orbit Model	356
29.11.5	Lorenz system	356
29.11.6	Maxwell's equations	357
29.11.7	Navier-Stokes equation for incompressible fluid	357
29.11.8	Predator Prey Model	357
29.11.9	Schrödinger's equation	357
29.11.10	Van der Pol oscillator	358
30	Expectation Maximization	359
30.1	The EM algorithm:	360
30.1.1	Motivation	361
30.2	Minorize-Maximize (MM) algorithms	361
30.3	Examples	363
30.3.1	Gaussian mixture	363
30.3.2	Blood Genotypes and Phenotypes	364
30.3.3	Missing Data/Imputation	366
31	Factor Analysis	367
32	Factor Graphs	371
33	Finite State Machine	374
33.1	Deterministic FSM	374
33.1.1	Example	374
33.1.2	Precise Definition	375
33.2	Non-deterministic FSM	376

33.2.1	Example	376
33.2.2	Precise Definition	377
33.3	Equivalency of deterministic and non-deterministic FSM	377
34	Frisch-Waugh-Lovell (FWL) theorem	379
34.1	FWL, assuming x^σ are non-random	379
34.2	FWL, assuming x^σ are random	380
35	Frontdoor Adjustment Formula	382
35.1	Examples	383
36	G-formula (Sequential Backdoor Adjustment Formula)	385
37	Gaussian Nodes with Linear Dependence on Parents	389
38	Gene Regulatory Networks	392
39	Generalized Linear Model (GLM)	394
39.1	Exponential Family of Distributions	394
39.2	GLM	396
40	Generative Adversarial Networks (GANs)	401
41	Goodness of Causal Fit	406
42	Gradient Descent	407
43	Granger Causality	409
44	Green's function	412
45	Hidden Markov Model	414
45.1	Calculating $P(x_t, v^n)$ and $P(x_t, x_{t+1}, v^n)$	416
45.2	Calculating $\mathcal{F}_t$ and $\overline{\mathcal{F}}_t$	417
45.3	Calculating $P(x^n v^n)$	418
45.4	Calculating $P(v^n A, B, \pi)$	419
45.5	Calculating $\hat{x}^n$ (Viterbi algorithm)	420
45.6	Calculating $\hat{A}, \hat{B}, \hat{\pi}$ (Baum-Welch algorithm)	422
46	Identification of do queries via LDEN diagrams	424
47	Influence Diagrams & Utility Nodes	426
47.1	Definitions	427
47.2	Variable Elimination Algorithm (VEA)	431

48	Instrumental Inequality and beyond	434
48.1	I-inequality	434
48.1.1	I-inequality for binary z,d,y	436
48.2	Bounds on Effect of IV on treatment outcome y	437
49	Instrumental Variables	440
49.1	δ with unmeasured confounder	440
49.2	δ (with unmeasured confounder) can be inferred via IV	441
49.3	More general bnets with IVs	442
49.4	Instrumental Inequality	443
50	Ising Dynamical Bayesian Network	444
50.1	Preliminaries	444
50.1.1	Information Theory	444
50.1.2	Standard Ising Model	446
50.2	Scaling a General Bnet	448
50.3	Scaling our Ising-dbnet Model	451
50.3.1	1-dim Ising bnet Model	451
50.3.2	2-dim Ising bnet Model	452
50.3.3	Converting 1-time bnet to a dynamical one	453
50.4	Software	453
51	Jackknife Resampling	457
51.1	Case $A = \Phi(x^n; n) = \frac{1}{n} \sum_{\sigma} x^{\sigma}$	459
52	Junction Tree Algorithm	461
53	Kalman Filter	462
53.1	Prediction Problem	463
53.2	Solution	464
53.3	Simple Example	465
53.4	Invariants	466
53.5	Derivation of Solution	466
54	Kernel Principal Component Analysis (Kernel PCA)	468
55	Kramers-Kronig Relations	472
55.1	Damped Harmonic Oscillator Example	476
56	LATE (Local Average Treatment Effect)	478
57	LDEN with feedback loops	484
58	Linear and Logistic Regression via grad descent	491

58.1	Generalization to x with multiple components (features)	493
58.2	Alternative $V(b, m)$ for logistic regression	493
59	Linear Deterministic Bnets with External Noise (LDEN Bnets)	495
59.1	Example of LDEN bnet	496
59.2	LDEN equations and their 2 solutions	497
59.3	Fully connected LDEN bnets	497
59.3.1	Fully connected LDEN bnet with $nx = 2$	498
59.3.2	Fully connected LDEN bnet with $nx = 3$	499
59.3.3	Fully connected LDEN bnet with arbitrary nx	502
59.4	Not fully connected LDEN bnets	503
59.5	LDEN bnet with conditioned nodes	504
59.6	SCuMpy	504
59.7	Non-linear DEN bnets	504
60	Marginalizer Nodes	506
61	Markov Blankets	508
62	Markov Chain Monte Carlo (MCMC)	510
62.1	Inverse Cumulative Sampling	510
62.2	Rejection Sampling	512
62.3	Metropolis-Hastings Sampling	513
62.4	Gibbs Sampling	516
62.5	Importance Sampling	518
63	Markov Chains	519
64	Mediation Analysis	520
65	Mendelian Randomization	527
66	Message Passing and Bethe Free Energy	529
66.1	2MRFs	529
66.2	Message Passing Intuition	530
66.3	$-\ln Z_\theta =$ Free Energy (FE)	534
66.4	$-\ln Z_{\theta^*} =$ Minimum FE	535
66.5	$-\ln Z_\theta^{tree} =$ Tree FE (a.k.a. Bethe FE)	536
66.6	$-\ln Z_{\theta^*}^{tree} =$ Tree Minimum FE, and message passing	537
67	Message Passing, Pearl's theory	540
67.1	Distributed Soldier Counting	540
67.2	Spring Systems	542
67.3	BP for Markov Chains	543

67.4	BP Algorithm for Polytrees	549
67.4.1	How BP algo for polytrees reduces to the BP algo for Markov chains	552
67.5	Derivation of BP Algorithm for Polytrees	553
67.6	Example of BP algo for a Tree	556
67.7	Bipartite bnets	560
67.8	BP for bipartite bnets (BP-BB)	561
67.8.1	BP-BB and general BP agree on Markov chains	563
67.8.2	BP-BB and general BP agree on tree bnets.	565
67.9	BP-BB and sum-product decomposition	566
68	Message Passing in Quantum Mechanics	568
69	Meta-learners for estimating ATE	569
70	Missing Data, Imputation	573
70.1	Imputation via EM	574
70.2	Imputation via MCMC	577
70.3	Multiple Imputations	578
71	Modified Treatment Policy	579
71.1	One time MTP	579
71.2	$\Delta_{ c}$ estimand	583
71.3	Estimates of $\Delta_{ c}$	586
71.3.1	Empirical estimate of $\Delta_{ c}$	586
71.3.2	OR estimate of $\Delta_{ c}$	586
71.4	Other Estimands besides $\Delta_{ c}$	589
71.5	Multi-time MTP	589
72	Monty Hall Problem	592
73	Multi-armed Bandits	594
73.1	Bnet for MAB	595
73.2	Reward functions	597
73.3	Regret functions	599
73.4	Strategies with random exploration	599
73.4.1	ϵ -greedy algorithm	600
73.4.2	ϵ_t -greedy algorithm	601
73.5	Strategies with nonrandom exploration	601
73.5.1	Upper Confidence Bounds (UCB) algorithms	601
	Frequentist UCB (UCB1) algorithm	601
	Bayesian UCB algorithm	602
73.5.2	Thompson Sampling MAB (TS-MAB) algorithm	603

Bnet for general TS-MAB algorithm	603
TS-MAB algorithm with Beta agent and Bernoulli environment	605
TS-MAB algorithm, skeletal reprise	606
73.5.3 Grad-MAB algorithm	607
74 Naive Bayes	610
75 Neural Networks	611
75.1 Activation Functions $\mathcal{A}_i^\lambda : \mathbb{R} \rightarrow \mathbb{R}$	612
75.2 Weight optimization via supervised training and gradient descent .	613
75.3 Non-dense layers	615
75.4 Autoencoder NN	617
76 Noisy-OR gate	618
76.1 3 ways to interpret the parameters π_i	619
77 Non-negative Matrix Factorization	622
77.1 Bnet interpretation	622
77.2 Simplest recursive algorithm	623
78 Observationally Equivalent DAGs	624
78.1 Examples	624
79 Omitted Variable Bias	627
80 Personalized Expected Utility	633
80.1 Goal of PEU Theory	634
80.2 Bnets for PEU Theory	635
80.3 Bounds on EU for unspecified bnet	635
80.4 Bounds on EU for specific bnet families	638
81 Personalized Treatment Effects	639
81.1 Goal, Strategy and Rationale of PTE theory	640
81.2 Bnets for PTE theory	642
81.3 $ATE = PB - PH$	643
81.4 Probabilities Relevant to PTE theory	644
81.5 Symmetry	649
81.6 Linear Programming Problem	650
81.7 Special constraints	651
81.8 Matrix representation of probabilities	653
81.9 Bounds on Exp. Probs. imposed by Obs. Probs.	657
81.10 Bounds on $PNS3$ for unspecified bnet	658
81.11 Bounds on $PNS3$ for specific bnet families	664
81.12 Bounds on ATE imposed by Obs. Probs.	664

81.13	Bounds on PNS in terms of ATE and Obs. Probs.	665
81.14	Numerical Examples	666
82	Petri Nets	668
82.1	Original Petri Net	669
82.1.1	Simple Example of Petri Net	669
82.1.2	Precise Definition of Petri Net	671
82.1.3	Firing of a Petri Net	672
82.2	Variants	673
82.2.1	Finite State Machine and Turing Machine	673
82.2.2	Continuous Petri Net	673
82.2.3	Colored Petri Net	674
82.2.4	Stochastic Petri Net	675
82.2.5	Bayes-Petri Net	675
83	Plate Notation	679
84	Potential Outcomes and Beyond	682
84.1	G and G_{den} bnets, the starting point bnets	683
84.2	G bnet with nodes $y^\sigma(0), y^\sigma(1)$ added to it.	685
84.3	Expected Values of treatment outcome y^σ	687
84.4	Translation Dictionary	688
84.5	$\mathcal{Y}_{ d,x} = \mathcal{Y}_{d d,x}$ (SUTVA)	689
84.6	Conditional Independence Assumption (CIA)	689
84.7	Treatment Effects	690
84.8	Insights into what makes treatment effects equal and $\mathcal{Y}_{1 0} = \mathcal{Y}_1$. . .	692
84.9	G_{do+} bnet	693
84.10	$ACE = ATE$	695
84.11	Good, Bad Controls	695
84.12	PO Confounder Sensitivity Analysis	697
84.13	Strata-Matching	699
84.13.1	Exact strata-matching	699
	Estimates of Treatment Effects	699
	Example, estimation of treatment effects	702
84.13.2	Approximate strata-matching	703
84.13.3	Unbiased strata-matching estimates	704
84.14	(SDO, ATE) space	705
84.15	Propensities	708
84.15.1	Propensity based estimates of Treatment Effects	710
84.15.2	Doubly Robust Estimates of Treatment Effects	712
84.15.3	Positivity	713
84.16	Multi-time PO bnets (Panel Data)	714

85	Principal Component Analysis	717
86	Program evaluation and review technique (PERT)	724
86.1	Example	726
87	Random Forest and Bagging	730
87.1	Bagging (with fully-featured bags)	730
87.2	Bagging (with randomly-shortened bags)	732
88	Recurrent Neural Networks	733
88.1	Language Sequence Modeling	736
88.2	Other types of RNN	736
88.2.1	Long Short Term Memory (LSTM) unit (1997)	738
88.2.2	Gated Recurrence Unit (GRU) (2014)	740
89	Regression Discontinuity Design	742
89.1	PO analysis	742
89.2	Linear Regression	744
90	Regularization of Loss Functions	745
90.1	L^p norm ROLF	747
90.1.1	L^1 norm ROLF can lead to sparsity	747
90.1.2	L^2 norm ROLF for Least Squares	749
90.2	Proximal functions	750
90.3	Proximal ROLF	752
90.4	Unobserved Nodes of a bnet	753
91	Reinforcement Learning (RL)	754
91.1	Exact RL bnet	757
91.2	Actor-Critic RL bnet	759
91.3	Q function learning RL bnet	761
92	Reliability Box Diagrams and Fault Tree Diagrams	763
92.1	Minimal Cut Sets	769
93	Renormalization Group, an Introduction	771
93.1	Introduction	771
93.2	Books and other references on RG	771
93.3	RG theory- a big tent	772
93.4	What is renormalization?	773
93.5	A whiff of thermodynamics	773
93.6	An essence of field theory	774
93.7	Naive versus fractal “scaling” dimensions	775
93.8	Real and momentum space RG	776

93.9	Correlations rule the world	777
93.10	Renormalization (semi-)group	777
93.11	RG streamlines	778
93.12	Fixed points of the trivial and critical kind	778
93.13	Critical exponents and universality classes	780
93.14	Relevant, marginal and irrelevant operators	781
93.15	Self-similar coupling constants and beta functions	782
93.16	The regulator and the fiducial mass scale	783
93.17	RG theory has its pi-groups too! Callan-Symanzik Type Equations	784
93.18	Forgetting initial conditions. Are we cheating with infinities? Where did the infinities go?	784
93.19	The many faces of a renormalizable theory	785
94	Restricted Boltzmann Machines	786
95	ROC curves	788
95.1	Terminology Table Adapted from Wikipedia Ref.[196]	791
96	Scoring the Nodes of a Learned Bnet	793
96.1	Probability Distributions and Special Functions	794
96.2	Single node with no parents	796
96.3	Multiple nodes with any number of parents	798
96.4	Bayesian Scores	800
96.5	Information Theoretic scores	800
97	Selection Bias Removal	802
97.1	Root and Leaf Nodes	802
97.1.1	R-L Conversion	803
97.1.2	R-L selector nodes	804
97.2	SB-Recoverability	805
97.2.1	Removing SB from passive query	805
97.2.2	Removing SB from active query	806
97.2.3	Examples	806
98	Sentence Splitting with SentenceAx	809
98.1	Preliminary Conventions	810
98.1.1	Tensor Notation	810
98.1.2	PyTorch conventions	811
98.2	Bayesian Network for this model	815
98.3	Loss for this model	817
99	Shannon Information Theory	820
99.1	Introduction	821

99.2	Preliminaries and Notation	822
99.3	Integration Over P-types	825
99.3.1	Integration Over Univariate P-type	825
99.3.2	Integration Over Multivariate P-types	828
99.3.3	Integration Over Conditional P-types	829
99.3.4	Dirac Delta Functions For P-type Integration	832
99.4	Source Coding (Lossy Compression)	833
99.5	Channel Coding	835
99.6	Source Coding With Distortion	844
99.7	Appendix: Some Integrals Over Polytopes	851
100	Shapley Explainability	857
100.0.1	Numerical examples of SHAP	860
101	Simpson's Paradox	863
101.1	Pearl Causality	865
101.2	Numerical Example	867
102	Stochastic Differential Equations	868
102.1	Notation	868
102.2	White Noise and Brownian Motion	869
102.3	SDE bnet	871
102.4	Simple Properties of SDE	873
102.4.1	STD with Constant Coefficients (CC)	874
102.4.2	Transition Probability Matrices	874
102.4.3	Markov chain	874
102.4.4	Chapman-Kolgomorov Equation	875
102.4.5	Martingale	875
102.5	Itô Integral	875
102.6	Fokker-Planck Equation	877
102.7	First and second order statistics	882
102.7.1	For general SDE	882
102.7.2	In case SDE has CC	885
102.8	Fourier Analysis for CC case	885
102.9	Lamperti Transformation	888
102.10	Feynman-Kac Path Integrals	889
102.11	Karhunen-Loève series	890
102.12	Girsamov Theorem	893
102.13	Doob's Transform	894
102.14	Appendix: Some explicitly solvable examples	897
102.15	Appendix: Ornstein-Uhlenbeck recurring example	898
103	Structure and Parameter Learning for Bnets	899

103.1	Overview	899
103.2	Score based SL algorithms	901
103.3	Constraint based SL algorithms	902
103.4	Pseudocode for some bnet learning algorithms	903
104	Support Vector Machines And Kernel Method	905
104.1	Learning Algorithm for SVM Classifier	906
104.2	Linear (dot-product) Kernel	907
104.3	Alternatives to Linear Kernel	910
104.4	Random Forest and Kernel Method	911
105	Survival Analysis	912
105.1	$S(t)$ estimates	914
105.1.1	No-censoring estimate of $S(t)$	914
105.1.2	Kaplan-Meier estimate of $S(t)$	914
105.2	$\lambda(t)$ models	919
105.2.1	$\lambda(t)$ independent of covariates Z	919
105.2.2	$\lambda(t)$ dependent on covariates Z	920
105.3	$S_0(t)$ estimates	923
106	Synthetic Controls	925
106.1	PO analysis	927
107	Table 2 Fallacy	929
108	Targeted Estimator	932
108.1	Goal, Strategy, and Rationale of TE theory	932
108.2	Functional Calculus	934
108.3	Linear Approximation of $\Psi[P_N]$	936
108.4	ATE estimand	937
108.5	ATE estimates	938
108.5.1	Ψ^E	938
108.5.2	Ψ^G	939
108.5.3	Ψ^{IPW}	939
108.5.4	Ψ^{LIPW}	939
108.5.5	Ψ^{LIPW++} (a.k.a. Ψ^{TMLE})	943
108.6	Ψ^{TMLE} in practice	945
109	Thermodynamics, a Causal Perspective	947
110	Time Series Analysis: ARMA and VAR	949
110.1	White noise	949
110.2	Backshift operator	950
110.3	Metrics	950

110.4	Definition of $ARMA(p, q)$, $AR(p)$ and $MA(q)$.	952
110.5	Solving $AR(p)$	954
110.6	Solving $MA(q)$	955
110.7	Solving $ARMA(p, q)$	956
110.8	Auto-correlation and partial auto-correlation	956
110.9	Generating function of auto-correlation	960
110.10	Impulse Response	961
110.11	$AR(p)$ and Yule-Walker equations	963
110.12	Forecasting	964
110.13	Model Learning	970
110.14	Differencing and $ARIMA(p, d, q)$	970
110.15	Parameter Learning	973
110.15.1	PL of $AR(p)$	974
110.15.2	PL of $MA(q)$	976
110.15.3	PL of $ARMA(p, q)$	978
110.16	$VAR(p)$	979
111	Transfer Learning	981
112	Transfer of Causal Knowledge	983
113	Transfer of Structural Knowledge	987
113.1	How to use Naive Bayes bnet	988
114	Transformer Networks	990
114.1	Tensor Notation	991
114.2	Recurrent Neural Net with Attention	992
114.2.1	Single Head Attention	992
114.2.2	Multi-Head Attention	995
114.3	Vanilla tranet	997
114.3.1	Single Head Attention	1002
114.3.2	Multi-Head Attention	1003
114.3.3	Encoder	1005
114.3.4	Decoder	1007
114.4	BERT	1009
114.4.1	BERT parameter values	1009
114.4.2	BERT Embedding	1010
114.4.3	BERT training	1011
115	Turbo Codes	1012
115.1	Decoding Algorithm	1015
115.2	Message Passing Interpretation of Decoding Algorithm	1017

116 Turing Machine	1018
116.1 Example	1019
116.2 Precise Definition	1020
117 Uplift Modelling	1021
117.0.1 Uplift definition	1022
117.0.2 Uplift2 definition	1023
117.0.3 UM customer types	1023
117.1 UM classifiers	1024
117.2 UM metrics and plots	1025
118 Variational Bayesian Approximation for Medical Diagnosis	1030
119 Variational Bayesian Approximation via D_{KL}	1034
119.1 Free Energy $\mathcal{F}(\vec{x})$	1036
120 XGBoost	1039
120.1 Divergences	1039
120.2 Minimizing Cost function for single tree	1041
120.3 Leaf Splitting	1044
120.4 Pruning	1044
120.5 Feature Binning	1046
120.6 Final estimate of target attribute	1046
120.7 Bnet for XGBoost	1047
121 YAML for bnet storage	1049
121.1 Getting acquainted with YAML	1050
121.2 Storing Bnets	1051
122 Zero Information Transmission (Graphoid Axioms)	1054
122.1 Consequences of Eq.(122.5)	1055
Bibliography	1057

Prologues

Chapter 50

Ising Dynamical Bayesian Network

In this chapter, we propose a dynamical Bayesian network (dbnet) (see Ch. 28) that is designed so as to incorporate all the physics of the standard Ising model (the latter is an undirected DAG (Markov model) instead of a DAG (bnet)). We have implemented our Ising-dbnet in the open source software `ising-dbnet` (see Ref.[97]).

This chapter and the software `ising-dbnet` are original research, first published here.

50.1 Preliminaries

In this section we remind the reader of some theory (on 1. Information theory 2. Standard Ising Model) that will be used to discuss the main topic of this chapter.

50.1.1 Information Theory

Recall that in Shannon Information theory (see Ch. 99), we define for random variables $\underline{x}, \underline{y}$

- the **entropy**

$$H(\underline{x}) = - \sum_{x \in \text{val}(\underline{x})} P(x) \ln P(x) \quad (50.1)$$

- the **joint entropy**

$$H(\underline{x}, \underline{y}) = - \sum_{x \in \text{val}(\underline{x})} \sum_{y \in \text{val}(\underline{y})} P(x, y) \ln P(x, y) \quad (50.2)$$

- the **conditional information**

$$H(\underline{y} \mid \underline{x}) = - \sum_{x \in \text{val}(\underline{x})} \sum_{y \in \text{val}(\underline{y})} P(x, y) \ln P(y \mid x) \quad (50.3)$$

$$= H(\underline{x}, \underline{y}) - H(\underline{x}) \quad (50.4)$$

- the **mutual information (MI)**

$$H(\underline{x} : \underline{y}) = \sum_{x \in \text{val}(\underline{x})} \sum_{y \in \text{val}(\underline{y})} P(x, y) \ln \frac{P(x, y)}{P(x)P(y)} \quad (50.5)$$

$$= H(\underline{y}) - H(\underline{y} \mid \underline{x}) = H(\underline{x}) - H(\underline{x} \mid \underline{y}) \quad (50.6)$$

Here are some standard inequalities that we will find useful in the rest of this chapter.

Claim 82

$$0 \leq H(\underline{x}) \leq \ln |\text{val}(\underline{x})| \quad (50.7)$$

$$0 \leq H(\underline{y} \mid \underline{x}) \leq H(\underline{x}, \underline{y}) \quad (50.8)$$

$$0 \leq H(\underline{x} : \underline{y}) \leq \min(H(\underline{x}), H(\underline{y})) \quad (50.9)$$

proof: Left to the reader.

QED

We will use MI to measure the correlation between random variables $\underline{x}$ and $\underline{y}$.

Note that MI achieves its zero lower bound when $P(x, y) = P(x)P(y)$ (i.e., $\underline{x}$ and $\underline{y}$ are uncorrelated).

Furthermore, MI achieves its upper bound given by Eq.(50.9) if either

- $\underline{y}$ is a deterministic function of $\underline{x}$; i.e.,

$$P(y \mid x) = \mathbb{1}(y = f(\underline{x})) \quad (50.10)$$

In that case, $\ln P(y \mid x)$ is either $0 \ln(0)$ or $\ln(1)$ so

$$H(\underline{y} \mid \underline{x}) = 0 \quad (50.11)$$

Furthermore,

$$H(\underline{y} \mid \underline{x}) = 0 \implies H(\underline{x} : \underline{y}) = H(\underline{y}) \quad (50.12)$$

- $\underline{x}$ is a deterministic function of $\underline{y}$; i.e.,

$$P(x \mid y) = \mathbb{1}(x = g(\underline{y})) \quad (50.13)$$

In that case, $\ln P(x \mid y)$ is either $0 \ln(0)$ or $\ln(1)$ so

$$H(\underline{x} \mid \underline{y}) = 0 \quad (50.14)$$

Furthermore

$$H(\underline{x} \mid \underline{y}) = 0 \implies H(\underline{x} : \underline{y}) = H(\underline{x}) \quad (50.15)$$

Define the **MI Efficiency** $\epsilon(\underline{y} \mid \underline{x})$ by

$$\epsilon(\underline{y} \mid \underline{x}) = \frac{H(\underline{y} : \underline{x})}{H(\underline{y})} = 1 - \frac{H(\underline{y} \mid \underline{x})}{H(\underline{y})} \quad (50.16)$$

From $0 \leq H(\underline{y} : \underline{x}) \leq H(\underline{y})$ we get

$$0 \leq \epsilon(\underline{y} \mid \underline{x}) \leq 1 \quad (50.17)$$

From its definition, we see that the MI efficiency has a singularity of the type $0/0$ when $H(\underline{y} \mid \underline{x}) = H(\underline{y}) = 0$.

We will henceforth refer to the $(H(\underline{y}), H(\underline{y} \mid \underline{x}))$ plane as the **entropy versus conditional info (E-CI) plane**

It is instructive (see Fig.50.1) to plot in the E-CI plane, the loci on which the efficiency equals certain values.

Another thing that I find instructive is to calculate the values of the efficiency in the case that $\underline{x}$ and $\underline{y}$ are simple binary random variables $\underline{x}, \underline{y} \in \{0, 1\}$. In that case, we represent $P(y|x)$ by a 2×2 matrix, with $\underline{y} \in \{0, 1\}$ labelling the rows and $\underline{x} \in \{0, 1\}$ labelling the columns. Fig.50.2) gives the values of the efficiency for various $P(y|x)$ and $P(x)$ possibilities.

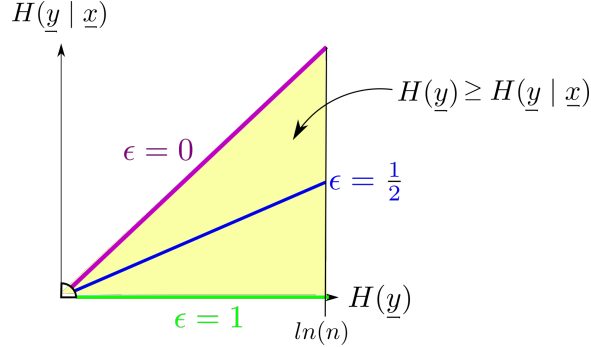


Figure 50.1: Efficiency $= \epsilon = \frac{H(\underline{y} : \underline{x})}{H(\underline{y})} = 1 - \frac{H(\underline{y} \mid \underline{x})}{H(\underline{y})}$. Efficiency has a singularity when both $H(\underline{y} \mid \underline{x})$ and $H(\underline{y})$ are zero.

50.1.2 Standard Ising Model

The standard Ising model is defined on a cubic lattice, as shown in Fig.50.3.

Let

$$Bool = \{0, 1\}$$

$$S_i \in \{-1, +1\} = 2Bool - 1 = \mathbb{Z}_{\pm}$$

T = temperature

$\beta = 1/(k_B T)$. We will set Boltzmann constant k_B to 1.

D = dimension of lattice. We refer to $D = 1$ as 1-dim, and to $D = 2$ as 2-dim.

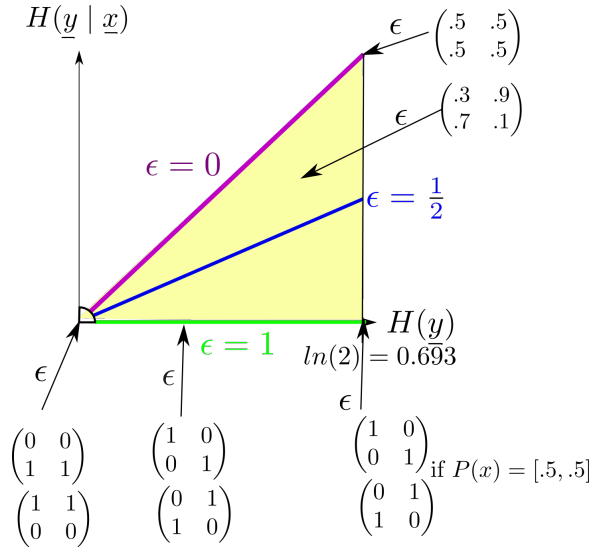


Figure 50.2: Efficiency for various values of the transition matrix $P(y | x)$ and the prior probabilities $P(x)$.

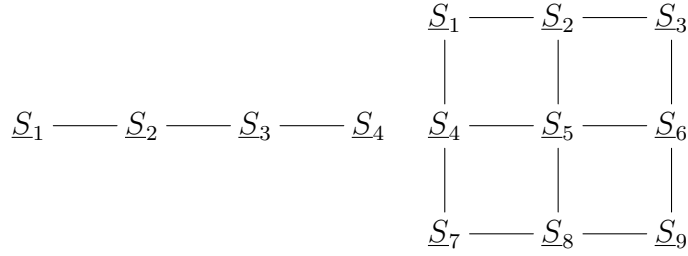


Figure 50.3: Ising Model cubic lattice for $D = 1$ (with 4 spins) and $D = 2$ (with 9 spins).

$NN(i)$ = set of indices of nearest neighbors to S_i .

$z = 2D$ = **coordination number**, number of nearest neighbors internally in the lattice, away from the boundary. $z = 2$ in $D = 1$ and $z = 4$ in $D = 2$. For spins on the boundary, the number of NN is less than z . For example, as you can see in Fig.50.3, for a rectangular lattice in 2-dim, the corner sites have 2 NN, and the non-corner boundary sites have 3 NN. Only the internal sites have 4 NN.

Assume coupling constants $J, h \geq 0$.

The **D-dim Ising Model** is defined as follows.

For $i = 1, 2, 3, \dots, n$, define **energies** $\mathcal{E}_i$ and $\mathcal{E}$ by

$$\mathcal{E}_i = \left[-JS_i \sum_{j \in NN(i)} S_j \right] - hS_i, \quad \mathcal{E} = \sum_{i=1}^n \mathcal{E}_i \quad (50.18)$$

and a Partition Function Z by

$$Z = \left[\prod_{i=1}^n \sum_{S_i \in \mathbb{Z}_{\pm}} \right] e^{-\beta \mathcal{E}} \quad (50.19)$$

Let

$\xi = \text{correlation length}$. $\langle \underline{S}_i, \underline{S}_{i+r} \rangle \sim e^{-r/\xi}$. $\xi \rightarrow \infty$ at critical points

$\langle m \rangle = \frac{1}{n} \sum_{i=1}^n \underline{S}_i = \text{magnetization}$

$T_c = 1/\beta_c = \text{Curie Temperature}$ (a.k.a. critical temperature). Magnetization is non zero for $T < T_c$ (spontaneous magnetization) and zero for $T > T_c$ (disordered phase).

The 2-dim Ising model with $h = 0$ has a critical point at $T = T_c$. If either $D = 1$ or $h \neq 0$, there are no critical points at any finite temperature, just phase transitions.

Ising Model can only be solved exactly for $D = 1, 2$. For other D , must solve numerically.

For the 2-dim Ising model with $h = 0$:

- Curie Temperature

$$T_c = \frac{2J}{\ln(1 + \sqrt{2})} \quad (50.20)$$

$$\beta_c J = \frac{1}{2} \ln(1 + \sqrt{2}) \approx 0.4407 \quad (50.21)$$

- correlation length

$$\xi \sim |T - T_c|^{-\nu}, \quad \nu = 1 \quad (50.22)$$

50.2 Scaling a General Bnet

Consider a general bnet with nodes $\underline{x}_i$ for $i = 1, 2, \dots, n$. Let

$\mathcal{R}$ = set of root nodes.

$\mathcal{R}^c$ = set of non-root nodes.

$x^{:n} = (x_1, x_2, \dots, x^n)$

Then

$$P(x^{:n}) = \left[\prod_{\underline{x}_i \in \mathcal{R}^c} P(x_i \mid pa(\underline{x}_i)) \right] \prod_{\underline{x}_i \in \mathcal{R}} P(x_i) \quad (50.23)$$

For $1 < b < \infty$, let

$$P_b(x_i | pa(\underline{x}_i)) = \frac{P(x_i | pa(\underline{x}_i))^b}{Z_i(b)} = \pi_i(b) \quad (50.24)$$

where

$$Z_i(b) = \sum_{x_i \in \text{val}(x_i)} P(x_i | pa(\underline{x}_i))^b \quad (50.25)$$

As $b \rightarrow \infty$,

$(\pi_1, \pi_2, \dots, \pi_n)$ tends to a one hot vector; i.e., all its components tend to zero except for one component, the one that is smallest at $b = 1$, which tends to 1:

$$\pi_i(b) \rightarrow \delta(i, i_0), \quad \text{where } i_0 = \underset{i}{\operatorname{argmin}} \pi_i(b = 1) \quad (50.26)$$

Therefore, $H(\underline{x}_i | pa(\underline{x}_i))$ (for P_b) tends to a sum of terms of the form $0 \ln(0)$ or $\ln(1)$. Thus

$$H(\underline{x}_i | pa(\underline{x}_i)) \rightarrow 0 \quad (50.27)$$

Thus

$$H(\underline{x}_i : pa(\underline{x}_i)) \rightarrow H(\underline{x}_i) \quad (50.28)$$

If $H(\underline{x}_i)$ remains finite as $b \rightarrow \infty$, then $\epsilon(\underline{x}_i | pa(\underline{x}_i)) \rightarrow 1$, but if $H(\underline{x}_i)$ tends to zero, then $\epsilon(\underline{x}_i | pa(\underline{x}_i))$ tends to a $0/0$ singularity at $H(\underline{y} | \underline{x}) = H(\underline{y}) = 0$

Claim 83

$$\sum_{i=1}^n H(\underline{x}_i) - H(\underline{x}^{:n}) = \sum_{\underline{x}_i \in \mathcal{R}^c} H(\underline{x}_i : pa(\underline{x}_i)) \quad (50.29)$$

Note that both sides equal zero when all the nodes are uncorrelated.

proof:

$$H(\underline{x}_i : pa(\underline{x}_i)) = \sum_{x_i} \sum_{pa(\underline{x}_i)} P(x_i, pa(\underline{x}_i)) \ln \frac{P(x_i | pa(\underline{x}_i))}{P(x_i)} \quad (50.30)$$

$$H(\underline{x}_i) = - \sum_{x_i} P(x_i) \ln P(x_i) \quad (50.31)$$

$$H(\underline{x}^{:n}) = - \sum_{x^{:n}} P(x^{:n}) \ln P(x^{:n}) \quad (50.32)$$

From Eq.(50.23), we get

$$\frac{P(x^{:n})}{\prod_{i=1}^n P(x_i)} = \prod_{\underline{x}_i \in \mathcal{R}^c} \frac{P(x_i \mid pa(\underline{x}_i))}{P(x_i)} \quad (50.33)$$

Hence

$$\sum_{x^{:n}} P(x^{:n}) \ln \frac{P(x^{:n})}{\prod_{i=1}^n P(x_i)} = \sum_{\underline{x}_i \in \mathcal{R}^c} H(\underline{x}_i : pa(\underline{x}_i)) \quad (50.34)$$

QED

This last claim says that the “bound entropy” (i.e., the entropy between when n random variables are independent and when they are correlated) equals the sum of the mutual informations for the non-root nodes.

Claim 84

$$\max_i H(\underline{x} : \underline{a}_i) \leq H(\underline{x} : \underline{a}_1, \underline{a}_2, \dots, \underline{a}_m) \leq \sum_{i=1}^m H(\underline{x} : \underline{a}_i) \quad (50.35)$$

proof:

We will prove this for $m = 3$ and leave the general case to the reader.

Averaging the logarithm of the following chain rule for conditional probabilities

$$\frac{P(x, a_1, a_2, a_3)}{P(x)P(a_1, a_2, a_3)} = \left[\frac{P(x, a_1 \mid a_2, a_3)}{P(x \mid a_2, a_3)P(a_1 \mid a_2, a_3)} \right] \left[\frac{P(x, a_2 \mid a_3)}{P(x \mid a_3)P(a_2 \mid a_3)} \right] \frac{P(x, a_3)}{P(x)P(a_3)} \quad (50.36)$$

we get

$$H(\underline{x} : \underline{a}_1, \underline{a}_2, \underline{a}_3) = H(\underline{x} : \underline{a}_1 \mid \underline{a}_2, \underline{a}_3) + H(\underline{x} : \underline{a}_2 \mid \underline{a}_3) + H(\underline{x} : \underline{a}_3) \quad (50.37)$$

Both sides of the inequality Eq.(50.35) follow from Eq.(50.37). Indeed,

$$H(\underline{x} : \underline{a}_1, \underline{a}_2, \underline{a}_3) = \underbrace{H(\underline{x} : \underline{a}_1 \mid \underline{a}_2, \underline{a}_3)}_{\leq H(\underline{x} : \underline{a}_1)} + \underbrace{H(\underline{x} : \underline{a}_2 \mid \underline{a}_3)}_{\leq H(\underline{x} : \underline{a}_2)} + H(\underline{x} : \underline{a}_3) \quad (50.38)$$

$\underbrace{\hspace{10em}}_{\geq H(\underline{x} : \underline{a}_3)}$

QED

Dividing Eq.(50.35) by $H(\underline{x})$ and identifying the $\underline{a}_i$ nodes with the parents $pa(\underline{x})$ of $\underline{x}$, we get

$$\max_{\underline{a} \in pa(\underline{x})} \epsilon(\underline{x} \mid \underline{a}) \leq \epsilon(\underline{x} \mid pa(\underline{x})) \leq \sum_{\underline{a} \in pa(\underline{x})} \epsilon(\underline{x} \mid \underline{a}) \quad (50.39)$$

50.3 Scaling our Ising-dbnet Model

So far, we have described a method of scaling any bnet. In this section, we define an Ising-dbnet and apply our scaling method to it. Our Ising-dbnet is designed to incorporate all the physics of the standard Ising model.

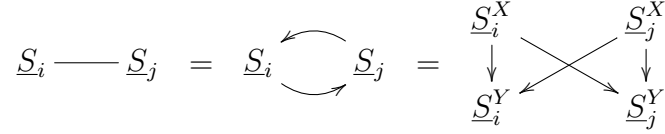


Figure 50.4: DAG definition of undirected arrow.

Let

$$Bool = \{0, 1\}$$

$$S_i, S_i^X, S_i^Y \in \{1, -1\} = 2Bool - 1 = \mathbb{Z}_\pm$$

$D_i = (S_i^X, S_i^Y)$ = “dipole node” (dnode) or “lattice site”

T = temperature

$$b = \beta = 1/T$$

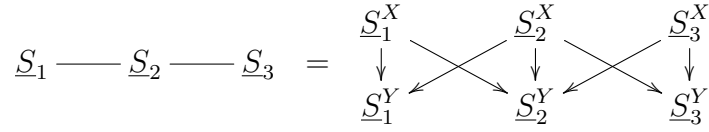


Figure 50.5: DAG definition of 1-dim Ising Model with 3 dnodes.

Assume coupling constants $J, h, \lambda \geq 0$. Actually, in our software `ising-dbnet` described later on, we only consider the case $h = \lambda = 0$.

50.3.1 1-dim Ising bnet Model

Our 1-dim Ising bnet model with 3 dnodes is defined as follows (generalization to n dnodes is obvious).

Consider the bnet of Fig.50.5 with node Transition Probability Matrices (TPM) given below, written in blue.

For $i = 1, 2, 3$,

$$P(S_i^X) = \frac{1}{2} \tag{50.40}$$

For $i = 2$,

$$\begin{cases} P(S_2^Y | S_2^X, S_1^X, S_3^X) = \frac{e^{-\beta \mathcal{E}_2}}{Z_2} \\ \mathcal{E}_2 = -JS_2^Y(S_1^X + S_3^X) - hS_2^Y - \lambda S_2^Y S_2^X \\ Z_2 = \sum_{S_2^Y \in \mathbb{Z}_\pm} e^{-\beta \mathcal{E}_2} \end{cases} \quad (50.41)$$

For $i = 1, 3$

$$\begin{cases} P(S_i^Y | S_i^X, S_2^X) = \frac{e^{-\beta \mathcal{E}_i}}{Z_i} \\ \mathcal{E}_i = -JS_i^Y(S_2^X) - hS_i^Y - \lambda S_i^Y S_i^X \\ Z_i = \sum_{S_i^Y \in \mathbb{Z}_\pm} e^{-\beta \mathcal{E}_i} \end{cases} \quad (50.42)$$

50.3.2 2-dim Ising bnet Model

Let

$\mathcal{B}$ = border dnodes of undirected graph

$\mathcal{B}^c$ = non-border, internal dnodes of undirected graph

n = number of nodes of undirected graph

Our 2-dim Ising bnet model with 9 dnodes is defined as follows (generalization to n dnodes is obvious).

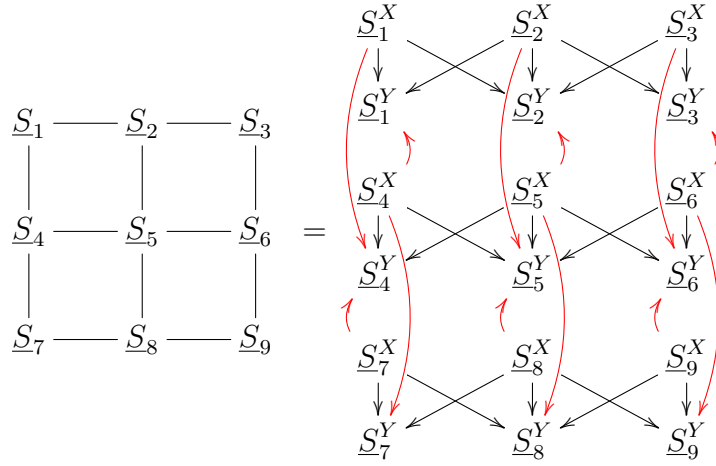


Figure 50.6: DAG definition of 2-dim Ising bnet with 9 dnodes on a 3×3 lattice.

Consider the bnet of Fig.50.6 with node Transition Probability Matrices (TPM) given below, written in blue.

For $i = 1, 2, \dots, n$

$$P(S_i^X) = \frac{1}{2} \quad (50.43)$$

For $\underline{S}_i \in \mathcal{B}^c$, if $\underline{S}_a, \underline{S}_b, \underline{S}_c, \underline{S}_d$ are the nearest neighbors (NN) of $\underline{S}_i$,

$$\begin{cases} P(S_i^Y | S_i^X, S_a^X, S_b^X, S_c^X, S_d^X) = \frac{e^{-\beta \mathcal{E}_i}}{Z_i} \\ \mathcal{E}_i = -JS_i^Y(S_i^X + S_a^X + S_b^X, S_c^X + S_d^X) - hS_i^Y - \lambda S_i^Y S_i^X \\ Z_i = \sum_{S_i^Y \in \mathbb{Z}_\pm} e^{-\beta \mathcal{E}_i} \end{cases} \quad (50.44)$$

For $\underline{S}_i \in \mathcal{B}$, the corner nodes have 2 NN and the non-corner ones have 3 NN. Eq.(50.44) is still valid for $\underline{S}_i \in \mathcal{B}$ if we set $\underline{S}_c = \underline{S}_d = 0$ in case of 2 NN and $\underline{S}_d = 0$ in case of 3 NN.

For the full bnet,

$$P(D^{:n}) = \left[\prod_{i=1}^n P(S_i^Y | pa(\underline{S}_i^Y)) \right] \prod_{i=1}^n P(S_i^X) \quad (50.45)$$

For a 2-dim Ising bnet, Claim 83 can be specialized as follows. Using $D_i = (S_i^X, S_i^Y)$ and the fact that the S_i^X nodes are all root nodes.

$$\underbrace{\sum_{i=1}^n H(\underline{x}_i)}_{\sum_{i=1}^n H(\underline{D}_i)} - \underbrace{H(\underline{x}^{:n})}_{H(\underline{D}^{:n})} = \underbrace{\sum_{\underline{x}_i \in \mathcal{R}^c} H(\underline{x}_i : pa(\underline{x}_i))}_{\sum_{i=1}^n H(\underline{S}_i^Y : pa(\underline{S}_i^Y))} \quad (50.46)$$

50.3.3 Converting 1-time bnet to a dynamical one

The 1-time Ising bnet presented so far does not equilibrate. To build a bnet that does equilibrate, we can make many copies (“times-slices”) of the above 1-time bnet and connect them to make a dynamic bnet.

Add a functional dependance (t) to all the random variables $\underline{D}_i$ in the 1-time bnet presented above. Do the same with $(t+1)$. Now we have two subsequent time slices.

To connect them, set $S_i^X(t+1)$ equal to $S_i^Y(t)$ for all i . In other words, we need to add the following TPM

$$P(S_i^X(t+1) | S_i^Y(t)) = \mathbb{1}(S_i^X(t+1) = S_i^Y(t)) \quad (50.47)$$

for all i .

50.4 Software

The ideas in this paper are implemented as free open source Python software at the github repo `ising-dbnet` (see Ref. [97])

Let

$\hat{\beta} = \beta/\beta_c$, where $\beta_c = 1/T_c$. $\hat{\beta} = 1$ at the Curie temperature T_c for a 2-dim Ising model.

In all our Jupyter notebooks, we take $J = 1$ and $h = \lambda = 0$

We consider a lattice that is 5×5 . At each lattice site $i = 1, 2, \dots, 25$, there are two nodes S_i^X and S_i^Y .

All the S_i^X nodes are root nodes of time slice t . The $S_a^X, S_b^X, S_c^X, S_d^X$ nodes of the nearest neighbors of site i point to S_i^Y

Iteration carries us from time slice t to time slice $t + 1$. At the beginning of iteration $t + 1$, we set $S_i^X(t + 1)$ equal to $S_i^Y(t)$ for all i .

p_0 is the initial ($t = 1$) probability $P(S_i^X(t = 1) = -1)$ for all i . A separate Jupyter notebook is provided for $p_0=0.3, 0.5, 0.7, p_0=\text{None}$, etc. ($p_0=\text{None}$ means each node is assigned a different p_0 sampled from the uniform($[0,1]$) probability distribution.

We present 3 types of plots

1. **lattice plots** (see Fig.50.7)

These show a 5×5 lattice with a color bar for efficiency ranging from 0 to 1.

Each of the 25 lattice sites has 2 arrows connecting it to each of its nearest neighbors.

The arrows are each labelled with the value of the efficiency for that arrow.

Sometimes the efficiency of an arrow is undefined. In that case, the arrow is dashed.

We use the probabilities $P(S_i^Y = \pm 1)$ to sample each node. If the sample turns out to be -1 (resp., $+1$), the node appears black (resp. white).

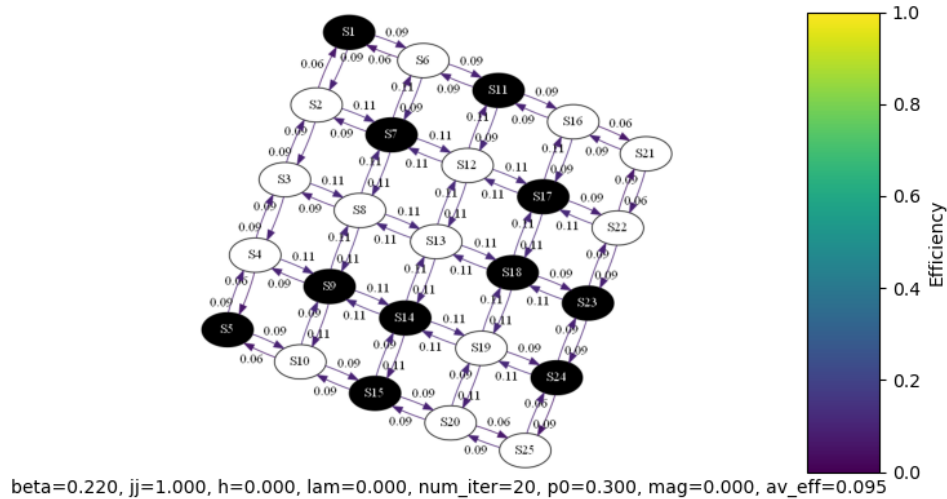


Figure 50.7: Example of lattice plot.

2. parametric plots (see Fig.50.8)

These show a parametric curve in the (x, y) plane, where

$$x = \text{average entropy} = \frac{1}{n} \sum_{i=1}^n H(S_i^Y)$$

$$y = \text{average conditional information} = \frac{1}{n} \sum_{i=1}^n H(S_i^Y | S_a^X, S_b^X, S_c^X, S_d^X)$$

$$n = 25$$

The parameter of the curves is $\hat{\beta}$.

The first and last point of the parametric curve are labelled by the value of the parameter $\hat{\beta}$.

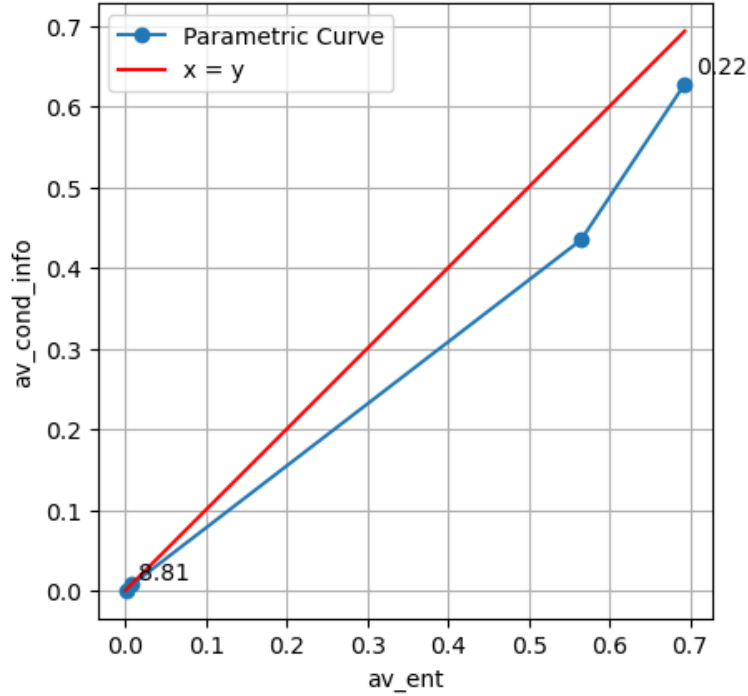


Figure 50.8: Example of parametric plot.

3. Magnetization plots(see Fig.50.9)

These show the magnetization $\text{mag} = \frac{1}{n} \sum_{i=1}^n S_i^Y$ as a function of $\text{beta_hat} = \hat{\beta}$

Conclusions

1. Our 2-dim Ising-dbnet model exhibits a discontinuity in the magnetization at the same Curie temperature as the standard vanilla 2-dim Ising model.
2. For $p_0 \neq .5$, the parametric curves flow smoothly as $\beta \rightarrow \infty$, towards $(H(y), H(y | x)) = (0, 0)$, where the efficiency is discontinuous.
4. For $p_0 = .5$, the parametric curves flow smoothly as $\beta \rightarrow \infty$, vertically down.

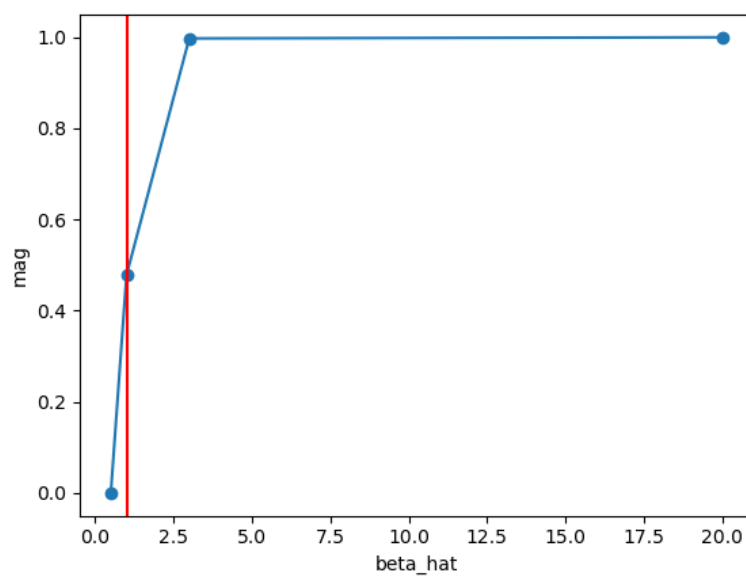


Figure 50.9: Example of magnetization plot.

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